

Ahmad Al-Imtiaz

Graduate Research Assistant, CASSA, Independent University, Bangladesh

Email: ahmadal.imtiaz@gmail.com | Website: <https://aaimtiaz.github.io>

EDUCATION

Master of Science in Physics [CGPA: 3.56 (out of 4)] Nov, 2024
Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

Bachelor of Science in Physics [CGPA: 3.28 (out of 4)] Oct, 2022
Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

RESEARCH INTERESTS

Observational Astronomy: Astronomical data analysis, photometric and spectroscopic techniques, and pipeline development. **Computational Astrophysics:** SED modeling, simulations, and computational methods for astronomical data. **Astrophysics:** Stellar populations and evolution, galaxy formation and evolution, chemical abundances, the circumgalactic medium (CGM), and gravitational lensing. My research interests are not limited to these areas; I am broadly interested in observational astronomy and astrophysics and open to exploring new research directions.

RESEARCH EXPERIENCE

Publications

- Adnan, S. M. R.[†], Hasan, M. J.[†], **Imtiaz, A. A.[†]**, Robin, S. H.[†], Shwadhin, F. R.[†], Shajib, A. J., Nahid, M. H., Tanver, M. H., Akter, T., Jahan, N., Jafar, Z., Rashid, M., Biswas, A., Chowdhury, A. A., Feardots, J., Rahaman, A., Ridwan, M., Sharma, R. D., Chowdhury, Z., Hossain, M. S.. [Investigating the relation between the environment and internal structure of massive elliptical galaxies using strong lensing](#) | *Astronomy & Astrophysics* [[†]these authors contributed equally to this work.]

Active Projects

Metallicity gradients estimation of the stellar populations on AGEL0014 In prep.
I investigated the stellar populations of the lensed quiescent galaxy AGEL0014 ($z \sim 1.4$). I reduced and analysed Gemini/GSAOI, HST, and Keck/MOSFIRE data, troubleshooting the [DRAGONS](#) pipeline by modifying local scripts to handle low source counts. I then performed multiband lens modelling and used [Lenstronomy](#)'s ray-shooting method to map image-plane segments to the source plane for spatially resolved spectral analysis. Finally, I performed SED fitting with [alf](#) and [Prospector](#) to measure metallicity and age gradients, providing insights into galaxy formation and quenching. Intensive computations were performed on the [NARIT HPC cluster](#). [A first-author paper is being prepared.]
Supervisor: Dr. Nicha Leethochawalit, NARIT, Thailand.

Co-evolution of the Star Clusters and the Host Galaxies across Cosmic Time In prep.
As a co-author in this CANUCS collaboration project, we are investigating the physical properties of compact stellar systems ("sparkles") within strongly lensed galaxies. My primary contribution was modeling the spectral energy distributions (SEDs) of the sparkles for the 43 galaxies from 5 different clusters. Using the [Dense Basis](#) framework, I have estimated both Simple Stellar Population (SSP) and complex stellar population properties.
Supervisor: Dr. Lamiya Mowla, Wellesley College, USA.

Automated Imaging Pipeline for the CASSA Observatory Expected: Dec 26
To prepare for first light at CASSA's upcoming 0.6 m optical observatory, I designed and developed an automated, Python-based [image reduction and photometry pipeline](#). Also, prepared another pipeline for atmospheric seeing monitoring ([DIMM](#)) with an 8" SkyWatcher setup on an EQ6-R Pro. These pipelines are in the draft phase and ready to test. After testing, further tweaking will be done.
Supervisor: Dr. Syed Ashraf Uddin, CASSA-IUB, Bangladesh.

SKILLS

- **Research Tools:** Mathematica, GitHub, Overleaf, ADQL, TOPCAT, APT, ds9, dragons (GSAOI data reduction pipeline), JWST data reduction pipelines, Lenstronomy, Dense Basis, alf, prospector
- **Programming language:** Python, C, MATLAB
- **Operating Systems:** Windows, Linux
- **Languages:** Bengali (Native), English (IELTS-6.5)
- **Telescopes:** Celestron CGEM II 800 SCT - 8" on CGEM II Computerized Equatorial Mount, EvoStar 80ED Apo Refractor on MEADE LX85 Computerized Equatorial Mount, Sky-Watcher Explorer-200P on an EQ6-R Pro Computerized Equatorial Mount

PROFESSIONAL & OTHER EXPERIENCES

- **Graduate Research Assistant**, CASSA, IUB (Full-time) [Contractual] Nov 2025–Present
I am collaborating on a Sparkler project with a PhD student from CUBoulder under Dr. Mowla's supervision. Additionally, I am building an [image reduction and photometry pipeline](#) for CASSA's first optical observatory, which aims to observe transient phenomena, especially supernovae, under the supervision of Dr. Uddin.
- **Project Facilitator and Consultant**, [EcoAstroBD](#), funded by IAU-OAD Dec 2025–Present
An initiative to tackle STEM inequality in Bangladesh by developing affordable, eco-friendly astronomy kits, where I provide consultation on leadership and coordination and facilitate organising activities.
- **Facilitator**, [TDMMA Workshop-2026](#), organized by CASSA-IUB 07-11 Sep 2026
Conducted a hands-on session on the [CASSA photometric pipeline](#).
- **Research Intern**, [ICTP PWF: Physics for Bangladesh Online Internships](#) Jul–Oct 2025
- **SOC & Instructor**, [CAM-SUST Summer School on Astrophysics 2025](#) Aug–Sep 2025
- **Instructor**, Stellar Light & Cosmic Bends: Data-Driven Astronomy Workshop 27–28 July 2025
Conducted a hands-on session on galaxy-galaxy strong lens modeling.
- **Lecturer**, [Dept. of CSE, Daffodil International University](#) (Full-time) [Contractual] Feb–Apr 2025
I taught PHY101 (Mechanics, Waves and Oscillations, Thermodynamics, Optics), PHY102 (Electricity and Magnetism, Modern Physics), and PHY 103 (Lab), which are junior undergraduate-level courses.
- **Asst. Teacher - Science** ([Reaz Public School](#)) (Full-time) [Permanent] Apr 2024–Feb 2025
Member of the [Corporate Social Responsibility team at Epyllion Group](#), Dhaka
- **Instructor**, AstroCode Intensive, CAM-SUST Mar 08–09, 2024
Conducted a hands-on session on astronomical image reduction techniques.
- **President**, 9th Executive Committee, [CAM-SUST](#) Mar 2021–Jul 2022
- **Study Circle Secretary**, 8th Executive Committee, [CAM-SUST](#) Aug 2019–Mar 2021

CONFERENCES

- "Resolved Metallicity and Age Gradients in the Lensed Quiescent Galaxy at $z \approx 1.4$ " Apr 09-11, 2026
Best oral presenter, International conference on Physics 2026, Bangladesh Physical Society, Bangladesh
- "[Exploring the Mass-Metallicity Relation in Open Clusters inside the Milky Way](#)" Nov 13–17, 2023
Poster at Metal Production and Distribution in a Hierarchical Universe - II, ESO Santiago, Chile
- "Stellar Multiplicity-dependent Neptunian Desert Boundary" Sep 23–26, 2022
Poster at International Conference of Physical Sciences by School of Physical Sciences, SUST

LEARNING EXPERIENCES

Schools and Workshops

- CASSA Workshop 1: [Unclogging the Thousand Eyes of an Array Radio Telescope](#), 2025 [In-person]
- [JWST Data Analysis and Processing Workshop \(South East Asia\)](#), 2024 [In-person], Organized by COSPAR & NARIT, Chiang Mai, Thailand
- [Astro101: Hands on Workshop on Optical Astronomy](#), 2023 (Hackathons Winner) [In-person], Organized by NARIT, Chiang Mai, Thailand
- [Introduction to Computational Astrophysics](#), 2021, NUEVACAMP, SPSB, MASLAB, Bangladesh
- [Introduction to HR Diagram with LCO](#), 2021, Prof. S N Hasan and Dr Priya Hasan, Maulana Azad National Urdu University, India
- [HPC for Astrophysics and Astronomy](#), 2021, SKA- India Consortium & IIT, Kharagpur, India
- [Introduction to MATLAB](#), 2021, IEEE NSU PES, Student Branch Chapter, Bangladesh
- [Introduction to Stars](#), 2021, Breakthrough Science Society, India
- [Machine Learning](#), 2021, Andrew Ng, Coursera
- 13th Radio Astronomy Winter School, 2020, IUCAA, Pune, India
- [Astronomical Data Analysis by TOPCAT](#), 2019, CAM-SUST, Sylhet, Bangladesh

COMPETITIONS

- [15th position in International Theoretical Physics Olympiad](#), Team Bosons, 2019 (Online)
- [Winner in University Physics Olympiad 2019](#), organized by CAM-SUST, Sylhet, Bangladesh (In-person)
- [Accomplished Competitor in The University Physics Competition](#), 2019 (Online)
- [Accomplished Competitor in The University Physics Competition](#), 2018 (Online)

OUTREACH

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| • Volunteer & consultant, CAM-SUST, Astro-Carnival, SUST (around 700 Students) | Sep 2023 |
| • Volunteer, CAM-SUST, Astronomy School Program, Jaflong (around 200 students) | Mar 2022 |
| • Convener, CAM-SUST, Astronomy School Program, Sunamganj (around 130 Students) | Mar 2020 |
| • Volunteer, CAM-SUST, Astronomy School Program, Sonargaon (around 150 students) | Jan 2020 |
| • Convener, CAM-SUST, Astronomy School Program, Brahmanbaria (around 500 Students) | Oct 2019 |
| • Volunteer, Physics Olympiad, National Round, BDPhO (around 700 Students) | Mar 2019 |

Oral Presentations

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| • Image Reduction on an off-line Study Circle of CAM-SUST | Jan 2024 |
| • Introduction to Stars II on an off-line Study Circle of CAM-SUST | Mar 2023 |
| • Introduction to Stars I on an online Study Circle of CAM-SUST | Oct 2022 |
| • Celestial Coordinate System on an online Study Circle of CAM-SUST | Oct 2020 |
| • Detection of Exoplanets, at a School Program Organized by CAM-SUST | Oct 2019 |

REFERENCES

Dr. Anowar J. Shajib
NFHP Einstein and KICP Fellow,
University of Chicago
ajshajib@uchicago.edu

Dr. Nicha Leethochawalit
Researcher,
National Astronomical Research
Institute of Thailand
nicha@narit.or.th

Dr. Syed Ashraf Uddin
Associate Professor, Department
of Physical Sciences, Independent
University, Bangladesh
uddinsa@iub.edu.bd